

EXERCISE I

In reference to the techniques used to reduce the computational effort in CNN

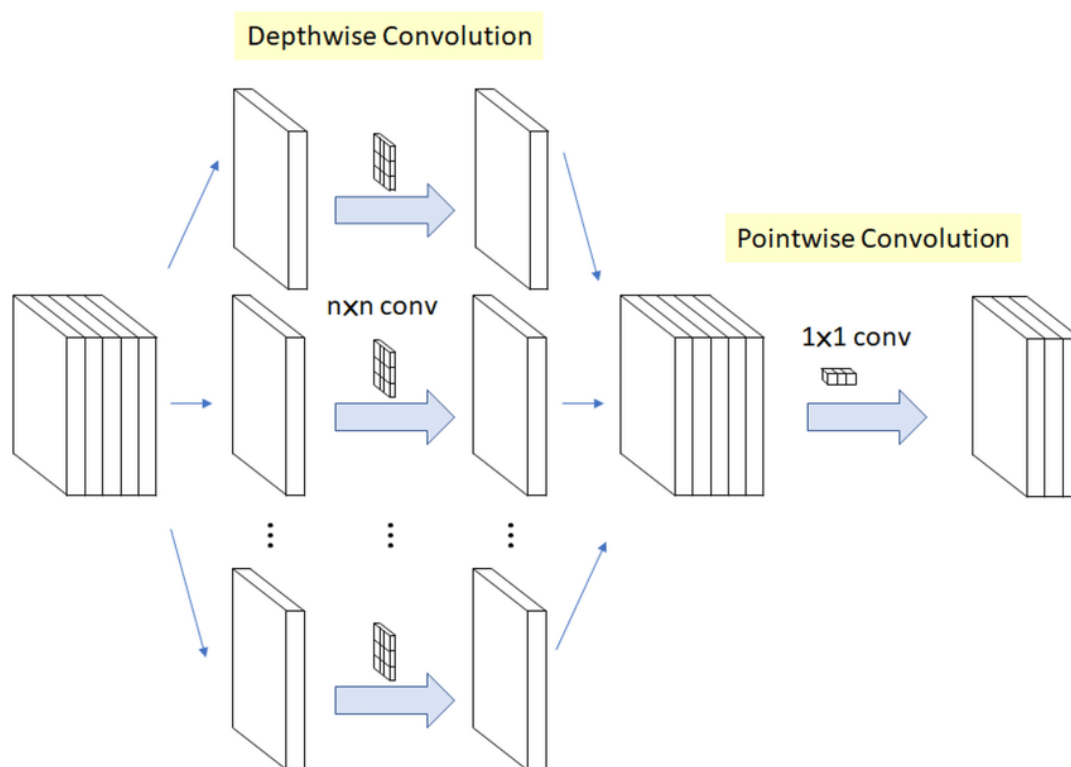
I.a) Describe the concept of depth-wise separable convolution.

<https://github.com/christianversloot/machine-learning-articles/blob/main/understanding-separable-convolutions.md>

A Separable Convolution is a technique to split a single convolution into two or more convolutions to produce the same output with less operations. In this way we reduce the memory and computational requirements of Conv2D layers.

In particular, for depthwise Separable convolutions the Conv2D layer are replaced with a sequence of a Depthwise convolution and a Pointwise Convolution.

- In the depthwise convolution each channel is considered separately, and one filter per channel is convolved over that channel only. This convolution acts on the spatial elements of the input activation while keeping the number of channels constant,
- The pointwise convolution is a normal 1x1 convolution. It varies the number of channels without altering the spatial size of its input activation.



I.b) Write a simple formula to express the number of MAC operations when you apply depth-wise convolution to N input channels ($C \times C$ size) and use a 5×5 filter size. Consider no padding and a stride of 1.

$K=5$ (kernel size)

$$C = W_{in} = H_{in}$$

$$\text{Output size} = (C + 2P - K) / (S) + 1 = C + 0 - 5 + 1 = C - 4$$

$$N^{\circ} \text{ MAC} = K * K * C_{in} * C_{out} * (W_{out} * H_{out}) \quad \text{MAC} = 5 * 5 * N * 1 * (C - 4) * (C - 4)$$

I.c) Describe the concept of pointwise convolution and compute the number of MACs and memory usage for weights (16 bits per weight) assuming that you have an input feature map of size 8x8 with 100 channels, and an output feature map with 128 channels

A Pointwise Convolution is a Conv2D with kernel size of 1.

$$\text{Weight_size} = K * K * C_{in} * C_{out} * \text{Nbytes} = 1 * 1 * 100 * 128 * 2 \text{ bytes} = 25600 \text{ B}$$

$$N^{\circ} \text{ MAC} = K * K * C_{in} * C_{out} * (W_{out} * H_{out}) \quad \text{MAC} = 1 * 1 * 100 * 128 * 8 * 8 = 819200 \text{ MAC}$$

EXERCISE II

Consider a convolutional neural network layer with the following characteristics: 64 input channels (22x22 spatial size), convolutional filters size 5x5, 100 output channels. Padding with zeros and stride 2.

II.a) Compute the number of MAC operations needed to compute a single output channel by applying convolution on a single input channel.

padding on a 5x5 kernel is 2 $\rightarrow P=2$

Stride: $S=2$

Spatial size = $W_{in} = H_{in} = 22$

Output size = $\text{floor}((W + 2P - K)/(S) + 1) = \text{floor}((22 + 2 \cdot 2 - 5)/2 + 1) = \text{floor}(8.5 + 1) = 11$

N° MAC 1 channel = $K \cdot K \cdot C_{in} \cdot C_{out} \cdot (W_{out} \cdot H_{out})$ MAC = $5 \cdot 5 \cdot 1 \cdot 1 \cdot 11 \cdot 11 = 3025$ MAC

II.b) Compute the total number of MAC operations needed to compute all the output channels in the layer.

N° MAC all channel = $K \cdot K \cdot C_{in} \cdot C_{out} \cdot (W_{out} \cdot H_{out})$ MAC = $5 \cdot 5 \cdot 64 \cdot 100 \cdot 11 \cdot 11 = 19360000$ MAC

II.c) Compute the number of SUM and number of DIV operations needed to perform average pooling (3x3) on all the output channels.

$N_SUM_1Ch = (3 \cdot 3 - 1) = 8$ SUM

$N_DIV_1Ch = 1$ DIV

$H_{out_pooling} = H_{out} - \text{Kernel_size_avg} + 1 = 11 - 3 + 1 = 9$

$W_{out_pooling} = W_{out} - \text{Kernel_size_avt} + 1 = 11 - 3 + 1 = 9$

$N_OPS = (N_SUM_1Ch + N_DIV_1Ch) \cdot C_{out} \cdot H_{avg_out} \cdot W_{avg_out}$
 $= 64800 \text{ SUM} + 8100 \text{ DIV}$

EXERCISE III

A parallel cluster of RISC-V processors is executing a convolutional layer with 17 input channels and 31 output channels. Input spatial size is 25x25. Inputs, outputs are quantized to 16 bits and weights are quantized to 8 bits. Convolutional filter size is 3x3, padding with zeros is used. Assume that on average a RISC-V processor can perform a MAC every 3 cycles. Assume that the computation is parallelized assigning output channels to processors using the following formula $PROCID = NL \% 8$ where $NL=0..30$ (ranges over the number of output channels).

III.a) Compute the L1 minimal shared memory size that is needed to store the complete layer (input and output channels as well as weights) in L1

act_bytes = 2 bytes, weight_bytes = 1 byte

Hin = Win = 25 (input spatial size)

Cin = 17

K = 3 (Kernel size)

P = 1 (padding)

Hout = Wout = 25 (input spatial size)

Cout = 31

in_size = Cin * Hin * Win = 17 * 25 * 25 = 10625

weights_size = K * K * Cin * Cout = 3 * 3 * 17 * 31 = 4743

out_size = Cout * Hout * Wout = 31 * 25 * 25 = 19375

L1_mem = (In_Size + Out_Size)*Act_bytes + weights_size*weights_bytes = 21250 + 38750 + 4743 = 64743 Bytes

III.b) Compute the total number of cycles to process the layer on a single processor

N_MAC = Hout * Wout * Cin * Cout * K * K = 25 * 25 * 17 * 31 * 3 * 3 = 2.964.375 MAC

Cycles (1 core) = N_MAC * 3 cycles = 8.893.125 cycles

III.c) Compute the speedup with respect to sequential when the cluster has 4 cores.

Assume 10% of stall cycles due to banking conflicts

Cycles_4cores_nostall = (Hout * Wout * Cin * ceil(Cout/Ncores) * K * K) * 3 cycles =

= (25 * 25 * 17 * 8 * 3 * 3) * 3 cycles = 2.295.000 cycles

Cycles_4cores = Cycles_4cores_nostall * 1.1 = 2.524.500 cycles

Speedup = Cyc_1C / Cyc_4C = 3.52 x